Compression Strategies for High-Throughput X-Ray CT of Natural History Collections

Jeppe Klitgaard[a,b,*], Carsten Gundlach[c,d], Christy Anna Hipsley[e,f], Mark D. Scherz[e], Kim Steenstrup Pedersen[e,g], Felipe Delestro Matos[a,b], Anders Bjorholm Dahl[a,b,c], Jakob Sauer Jørgensen[a,b]

[a]*Center for Quantification of Imaging Data from MAX IV, Kongens Lyngby, Denmark*
[b]*Department of Applied Mathematics and Computer Science, Technical University of Denmark, Kongens Lyngby, Denmark*
[c]*3D Imaging Centre, Technical University of Denmark, Kongens Lyngby, Denmark*
[d]*Department of Physics, Technical University of Denmark, Kongens Lyngby, Denmark*
[e]*Natural History Museum of Denmark, University of Copenhagen, Copenhagen, Denmark*
[f]*Department of Biology, University of Copenhagen, Copenhagen, Denmark*
[g]*Department of Computer Science, University of Copenhagen, Copenhagen, Denmark*

The NaturArv3D project[1] will acquire 3D X-ray images of the full vertebrate collection of the Natural History Museum of Denmark at sub-10 µm voxel size, in collaboration with the DTU 3D Imaging Centre. The scale of the project necessitates a custom high-throughput X-ray scanner with robotic sample handling, capable of scanning up to 500 samples per week. The expected annual data volume of approximately 1 PB renders long-term archival storage economically onerous without substantial compression.

The compression ratios (CR), compression and decompression times, and quality loss are determined for in-memory compression tests across a large number of lossless and lossy compression schemes, including JPEG-LS and the recent JPEG XL standard. Relevant subsets of the tests are performed in TIFF, DICOM, and (OME-)Zarr containers in order to determine the most viable archival formats and their compatibility with prevalent image-processing software packages. Quality loss is quantified using common full-reference metrics including SSIM [1], MS-SSIM [2], and PSNR.

We find that lossless schemes yield a maximum CR of 1.75:1, corresponding to a 43% space saving (SS). The JPEG-LS and JPEG XL codecs achieve CRs of 16.3:1 (93.9% SS) and 45.6:1 (97.8% SS), respectively, at imperceptible quality loss ($\mathrm{PSNR} \approx 60\,\mathrm{dB\ full\ scale}, \mathrm{3D\ SSIM} \approx 0.999$) on a representative 16-bit CT volume. Alongside JPEG 2000, these schemes in combination with a DICOM container constitute the only lossy options widely supported among the surveyed packages. Compression times remain compatible with high-throughput acquisition, while decompression is sufficiently fast to improve access times over the uncompressed, I/O-bound case.

Relative to the baseline data pipeline without compression, the proposed schemes offer an estimated annual storage-cost saving of 2 million DKK under realistic operating conditions. This underscores that the adoption of lossy compression is decisive for the economic viability of high-throughput X-ray imaging at scale.

References

[1] Z. Wang, A. Bovik, H. Sheikh, and E. Simoncelli, "Image Quality Assessment: From Error Visibility to Structural Similarity," *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, Apr. 2004, doi: 10.1109/TIP.2003.819861.

[2] Z. Wang, E. Simoncelli, and A. Bovik, "Multiscale Structural Similarity for Image Quality Assessment," in *The Thrity-Seventh Asilomar Conference on Signals, Systems & Computers, 2003*, Nov. 2003, p. 1398–1402 Vol.2. doi: 10.1109/ACSSC.2003.1292216.

[*]E-Mail: s250250@dtu.dk

[1]https://3dim.dtu.dk/newsarchive/2025/03/naturarv_3d